

English proficiency: exploring future outlooks, and to what extent expected covariate-based trends occur across Italian regions and the EU - **Project proposal**

Project idea description

Topic

This project aims to inform policy decisions aimed at positively affecting English proficiency across Italy. Proficiency in English is considered by 77% of Europeans as “crucial for personal development” (European Union, 2024a, p. 92), and the probability of being employed for Italians in Italy significantly increases if they speak English (by 5.7% for women and by 4.3% for men) (Gazzola and Mazzacani, 2019), without counting all the many Italians who move abroad (Istituto Nazionale di Statistica, 2025). Notwithstanding, Italy ranked 35th out of the 37 European countries in the latest Education First (EF) English Proficiency Index (EPI) (Education First, 2025a). Some data-driven overviews are in order.

Firstly, while research exists that exposes some variables’ associations with English proficiency, with English proficiency being either a potential outcome (e.g., education variables) or a potential predictor (e.g., economic variables) (Sorrenti, 2017; De Wilde, Brysbaert and Eyckmans, 2019; Garrett, Davis and Eisner, 2019; Education First, 2025a), this research has typically had different scopes (rather than being specific to Italy) or scales (rather than national or regional scales). Through issues that in computational social science are described as transportability problems and scale-up problems, this limits inference for policy purposes. Testing and subsequent visualisations are in order at the regional level across Italy, or, where no reliable data exists for the latter, at the national level across Europe, including Italy. This is to understand not the mechanisms of these trends, but whether these trends seem to hold at the scales certain policymakers work with, and their magnitude at these scales, though without the benefit experimental designs have for causal inference.

Secondly, while data exists that has at least a small chance to detect immediate effects of upper-secondary-school English as a second language (ESL) across European countries (including Italy) on young adults’ English proficiency (Education First, 2026; European Union, 2026a), such trends need to be tested and visualised.

Thirdly, while data exists on the state of ESL teaching within Italy per region (INVALSI Statistical Department, 2025; 2026a; 2026b; Ministero dell'Istruzione e del Merito, 2023; 2024), distinct parts of it originate from distinct sources, although they can be aggregated in a meaningful way.

Thus, it is in order to visualise data from a number of sources through which meaningful questions, such as those in this proposal's next subsection, can be answered.

Circumstances and purpose

The primary audience is policymakers, though economists and other researchers are expected to find the visualisations relevant too. No scientific background will be assumed: technical terms will be either explained in context or replaced with plainer language, keeping the format accessible and explanatory throughout. The project's main purpose is to inform, providing data-driven answers to several interrelated questions. For example: how noticeably ESL exposure relates to later EF English proficiency statistics for adults across EU countries (though with measurement-related constraints that future research may choose to minimise); associations with economic indicators that English proficiency tends to have specifically across Italian regions; whether incentivising ESL teachers to relocate to specific Italian regions could be a worthwhile policy investment; and how English proficiency across Italian regions is associated differently with different candidate predictors such as internet use, schooling, and Erasmus participation (while acknowledging that these associations may be heavily confounded). The takeaway should be an evidence-grounded picture of where English proficiency stands across Europe and Italy, some variables it correlates with, and where policy levers may or may not plausibly lie.

The project will be presented as an interactive dashboard (Plotly) combining maps and comparative views. It will be designed for web-based exploration, allowing users to navigate across countries, variables, and time.

Project data and project definition

All data whose use for the project is planned is publicly available. Some data is stored in formats that may require manual transcription — this will not be a problem, because the visualisation will not be updated automatically with time. For the three parts into which the visualisations will be divided, the data will be as follows.

Data requirements

Key variables include EF EPI scores (national and regional), ESL exposure rates (share of pupils learning English), and selected covariates such as internet use, Erasmus participation, and regional indicators of development (HDI, RCI, SPI). These variables will be harmonised across datasets and, where necessary, standardised to allow meaningful comparisons across countries and regions.

Highlighting key features

Key insights will be highlighted through the use of color gradients in choropleth maps, regression lines in scatter plots, and simple interactive filtering (e.g., by year or variable). Particular attention will be given to deviations from general trends, in order to make cross-country differences and regional disparities more immediately visible.

Part 1 of 3: Are trends in Europe involving upper-secondary ESL education and adult self-tests for English proficiency the trends we would expect?

Data

Education First English Proficiency Index (EF EPI): nation-level data (Education First, 2026). Data from 2016 to 2025. Key limitation: bias — each year's report has results from tests taken at home by people who went on the EF website presumably in order to test their own English proficiency. It is assumed that whatever bias this introduces does not fatally harm content validity and is constant enough across regions and nations to make it negligible for comparisons.

“Pupils by education level and modern foreign language studied - absolute numbers and % of pupils by language studied” (European Union, 2026a).

Sub-project definition

This sub-project will evaluate whether expected temporal relationships between exposure to English as a second language (ESL) in upper-secondary education and later adult English proficiency are observable across EU countries.

To account for delayed effects, ESL exposure and EF EPI scores will be aligned using a temporal lag (t for ESL, $t+4$ for EF EPI) and standardised (z-scores) to ensure comparability across countries and years.

The analysis will rely on three main visualisations (with a year slider for the user to choose the year for t): (i) an animated choropleth map of EF EPI scores (2016–2025) to capture spatial and temporal patterns, (ii) a dynamic scatter plot with individual countries as data points, relating ESL

exposure to later proficiency (x = number of upper-secondary students learning English that year, y = EF EPI score four years later), where the dot for Italy will be highlighted, plus a regression line, and (iii) country-level time series highlighting selected cases (e.g., Italy).

These views aim to assess whether higher ESL exposure is associated with subsequent improvements in English proficiency, while identifying cross-country variations and deviations from general trends. Given the observational nature and known biases of EF EPI data, the focus remains on patterns and associations rather than causal inference.

Part 2 of 3: Are correlates of English proficiency across Italian regions those we would expect, and how strong are they?

Data

One data point per variable per “NUTS 2” region (Italian regions, and regions within all other EU countries).

Education First English Proficiency Index (EF EPI): region-level data in Italy (Education First, 2023; 2024; 2025b). Data from 2023. Earlier years are unavailable, foreclosing a more comprehensive project, but this data is sufficient for the purposes of part 2 of 3 described below. This issue adds to the issues of the EF EPI mentioned in part 1 of 3 above, which, as mentioned there, may be negligible for these purposes.

Potential predictors of EF EPI: Erasmus (European Union, 2024b) and Internet use (European Union, 2026b).

Other correlates of EF EPI: Regional Competitiveness Index (RCI) (European Union, 2022), Social Progress Index (SPI) (European Union, 2024c), Human Development Index (HDI) (Global Data Lab, 2026). For the first two, data are not gathered annually, but are data from late 2010s to early 2020s gathered into one data point per region. However, they will be treated on a par with the most recent data from the HDI, even though the latter is gathered annually per region. This is because simply the most recent data point from each index will be considered for a valid comparison, and the ensuing issue is essentially only measurement precision varying across indices. The fact that, by contrast, the candidate predictor (EF EPI) is, as mentioned, from 2023 means that causality is bound to remain highly unclear and partially bound by whether EF EPI 2023 data for individual Italian regions is a proxy for English proficiency within those regions even some years prior, and even just before or early into the pandemic. Otherwise, causality cannot be hypothesised due to the effect coming at least partly before the cause.

Sub-project definition

This sub-project will statistically test these trends across NUTS 2 regions across the EU (test results only will be presented), and visualised across NUTS 2 regions of Italy only. The test is wider for more statistical power, though with a transportability cost when making inferences for Italy only, even though the visualisation will involve the data gathered for Italy only.

Comparing EF EPI English proficiency with Erasmus data, internet use data, RCI, SPI, and HDI will result in five comparisons. For each of these, the plan is to make one choropleth map for English proficiency, and one for the other variable, displayed side-by-side. An additional choropleth map will show the difference magnitude between them (standardised), and lastly, there will be a scatter plot whose regression line will be highlighted if the test across the EU is significant.

Part 3 of 3: Future prospects across Italian regions: INVALSI for English proficiency and pupils per English teacher

Data

One data point per variable per Italian region.

Latest English-teaching quality per region as reported by the INVALSI Statistical Department (2025, 2026a, 2026b).

Latest computable ratio of pupils per English teacher within each region. These statistics will be computed by us from two sources from the Ministero dell'Istruzione e del Merito (2023; 2024).

Sub-project definition

This sub-project is essentially descriptive. One static choropleth map will be made for each of the two variables. The two maps will be displayed side-by-side and with the same colour scale.

Format and medium

Each of the web-based interactive dashboard's three unordered sections will correspond to one of the three parts described above. Because different users will have different questions, no strict order for the sections will be suggested or imposed. To avoid cluttering, no visual elements are planned other than those described so far and those needed to navigate the dashboard.

Expected challenges

One key challenge may be 2020's COVID-19 disruption (at least for Part 1). To address it, standardisation across measurements in each given year can be put in place to reduce pandemic-related noise, and make any remaining anomalous patterns more identifiable.

Additionally, data availability and processing represent a practical challenge. While the project relies on open-access datasets, these are often heterogeneous, partially incomplete, or not directly usable. This will require data extraction (e.g., from PDFs), harmonisation across sources, and preprocessing.

More broadly, inference remains limited, as the observed patterns are multi-causal and the project primarily focuses on visualisation. However, it remains relevant as it provides a coherent and comparative overview of trends.

Reference list

De Wilde, V., Brysbaert, M., and Eyckmans, J. (2020) 'Learning English through out-of-school exposure. Which levels of language proficiency are attained and which types of input are important?', *Bilingualism: Language and Cognition*, 23, pp. 171–185. Available at: <https://doi.org/10.1017/S1366728918001062>

Education First (2023) *EF EPI 2023 – EF English Proficiency Index Factsheet – Italy*. Available at: <https://www.ef.com/assetscdn/WIBlwq6RdJvcD9bc8RMd/cefcom-epi-site/fact-sheets/2023/ef-epi-fact-sheet-italy-english.pdf> (Accessed: 16 March 2026).

Education First (2024) *Italy | EF English Proficiency Index | EF Global Site (English)*. Available at: <https://www.ef.com/assetscdn/WIBlwq6RdJvcD9bc8RMd/cefcom-epi-site/fact-sheets/2024/ef-epi-fact-sheet-italy-english.pdf> (Accessed: 16 March 2026).

Education First (2025a) *EF English Proficiency Index*. Available at: <https://www.ef.com/assetscdn/WIBlwq6RdJvcD9bc8RMd/cefcom-epi-site/reports/2025/ef-epi-2025-english.pdf> (Accessed: 10 March 2026).

Education First (2025b) *Italy | EF English Proficiency Index | EF Global Site (English)*. Available at: <https://www.ef.com/assetscdn/WIBlwq6RdJvcD9bc8RMd/cefcom-epi-site/fact-sheets/2025/ef-epi-fact-sheet-italy-english.pdf> (Accessed: 16 March 2026).

Education First (2026) *About EF EPI*. Available at: <https://www.ef.com/wwen/epi/about-epi/> (Accessed: 7 April 2026).

European Union (2022) *EU Regional Competitiveness Index 2.0 - 2022 edition*. Available at: https://ec.europa.eu/regional_policy/assets/regional-competitiveness/index.html#/ (Accessed: 10 April 2026).

European Union (2024a) *Europeans and their languages*. Available at: <https://doi.org/10.2766/28257>

European Union (2024b) *Erasmus+ Mobility Raw Data*. Available at: <https://data.europa.eu/data/datasets/erasmus-mobility-raw-data?locale=en> (Accessed: 4 April 2026).

European Union (2024c) EU Social Progress Index 2.0 - 2024 edition. Available at: https://ec.europa.eu/regional_policy/assets/social-progress/index.html#/ (Accessed: 10 March 2026).

European Union (2026a) *Pupils by education level and modern foreign language studied - absolute numbers and % of pupils by language studied*. Available at: https://doi.org/10.2908/EDUC_UOE_LANG01

European Union (2026b) Internet use by individuals. Available at: <https://doi.org/10.2908/TIN00028>

Garrett, R., Davis, E. and Eisner, R. (2019) 'Student and school characteristics associated with academic performance and English language proficiency among English learner students in grades 3–8 in the Cleveland Metropolitan School District', *American Institutes for Research*. Available at: https://ies.ed.gov/sites/default/files/migrated/rel/regions/midwest/pdf/REL_2019003.pdf (Accessed: 10 March 2026).

Gazzola, M. and Mazzacani, D. (2019) 'Foreign language skills and employment status of European natives: evidence from Germany, Italy and Spain', *Empirica*, 46, pp. 713-740. Available at: <https://doi.org/10.1007/s10663-019-09460-7>

Global Data Lab (2026) *Subnational HDI (v10.1)*. Available at: <https://globaldatalab.org/shdi/table/shdi/ALB+AND+AUT+AZE+BLR+BEL+BIH+BGR+HRV+CYP+CZ+E+DNK+EST+FIN+FRA+DEU+GRC+HUN+ISL+IRL+ITA+XKO+LVA+LIE+LTU+LUX+MDA+MNE+NLD+MKD+NOR+POL+PRT+ROU+RUS+SRB+SVK+SVN+ESP+SWE+CHE+UKR+GBR/?levels=1+4&interpolation=0&extrapolation=0> (Accessed: 5 April 2026).

INVALSI (2025) *Micro-data Sample G05 2024-25*. Available at: https://serviziostatistico.invalsi.it/en/invalsi_ss_data/microdati-campione-g05-2024-25/ (Accessed: 4 April 2026).

INVALSI (2026a) *Microdati Campione G08 2024-25*. Available at: https://serviziostatistico.invalsi.it/en/invalsi_ss_data/microdati-campione-g08-2024-25/ (Accessed: 4 April 2026).

INVALSI (2026b) *Microdati Campione G13 2024-25*. Available at: https://serviziostatistico.invalsi.it/en/invalsi_ss_data/microdati-campione-g13-2024-25/ (Accessed: 4 April 2026).

Istituto Nazionale di Statistica (2026) *Demographic Indicators – Year 2025*. Available at: <https://www.istat.it/en/press-release/demographic-indicators-year-2025/> (Accessed: 7 April 2026).

Ministero dell'Istruzione e del Merito (2023) *Portale unico dei dati della scuola*. Available at: <https://dati.istruzione.it/opensdata/> (Accessed: 10 March 2026).

Ministero dell'Istruzione e del Merito (2024) *Focus “Principali dati della scuola – Avvio Anno Scolastico 2024/2025”*. Available at: https://www.mim.gov.it/web/guest/pubblicazioni/-/asset_publisher/6Ya1FS4E4QJw/content/focus-principali-dati-della-scuola-avvio-anno-scolastico-2024-2025- (Accessed: 10 March 2026).

Sorrenti, G. (2017) 'The Spanish or the German apartment? Study abroad and the acquisition of permanent skills', *Economics of Education Review*, 60, pp. 142-158. Available at: <http://dx.doi.org/10.1016/j.econedurev.2017.07.001>